Clamshell Device - Block Diagram (Initial)

Assumptions: ESP32-S3 main module, two 2.4" MIPI-DSI AMOLEDs, USB-C PD charging (BQ25895), single thin Li-Po battery, rigid-flex PCB (two rigid halves + flex hinge).

Blocks:
- ESP32-S3 Module (GPIOs, MIPI DSI lanes, SPI/I2C for touch, UART for debug)
- Display 1 (left half) : 2.4" MIPI-DSI AMOLED (on-cell touch if available)
- Display 2 (right half): 2.4" MIPI-DSI AMOLED (on-cell touch if available)
- Power Management: BQ25895 + USB-C CC (STUSB4500) + battery protection + fuel-gauge
- USB-C Connector for charging and UART (bootloader/programming)
- Sensors: Hall sensor for lid detect, Alert button (capacitive or tactile)
- Antennas: WiFi/BT antenna on one edge; provision for cellular antenna omitted

Interconnections:
- MIPI-DSI lanes from ESP32-S3 to each display (use FFC or flexible trace across hinge)
- I2C for touch controller (or internal on-cell), ADC for battery sense, UART for debug
- Power rails (VBUS, battery, 3.3V) routed with power planes and decoupling